

Phase-space contraction and attractors for ultrarelativistic electrons

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Ultrashort-pulse laser intensities can reach 10^{22} W/cm². In this case the electron motion becomes ultrarelativistic and significant bremsstrahlung occurs. The radiation causes a dissipative effect, which is called a radiation reaction. It has been shown in the literature that the radiation reaction force causes phase-space contraction when the motion of electrons in a laser field is considered. The effect of the radiation reaction force is smaller for electron propagation in the direction of a propagating plane wave compared to counterpropagation. In the case of two colliding laser beams with sufficiently large amplitudes, stochastic heating is an important process that will be influenced by the radiation reaction. It is shown here that the radiation reaction causes attractors and in certain parameter regimes electron motion converges to regular attractors. This causes a significant reduction of stochastic heating. The forms of the attractors are presented. The results confirm the general prediction of phase-space contraction and provide quantitative results for the speed of contraction as well as the type of attracted motion.

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I. INTRODUCTION

The ultrarelativistic laser-electron interaction is an active field of research [1]. Ponderomotive laser acceleration [2,3] is one of the fascinating applications in the field. Laser intensities of 10^{19} Watt/cm² are now available in terawatt tabletop laser systems based on chirped-pulse amplification. Large laser systems may deliver intensities up to $I \approx 10^{23}$ W/cm² at the focal spot. In the near future, extreme laser intensities $I > 10^{23}$ W/cm², when the electrons become ultrarelativistic within a fraction of a wave period, are expected to become available. The electron motion will be accompanied by significant bremsstrahlung. When the corresponding radiation reaction (friction) force becomes of the same magnitude as the Lorentz force one enters the so-called radiation-dominated regime [4,5]. An accelerated electron will emit quite a large amount of electromagnetic radiation and the emitted field will influence the motion of the electron itself, an effect that is covered by the radiation reaction (RR) force [6,7]. The inclusion of RR effects in a classical equation of motion goes back to Lorentz, Abraham, and Dirac; see, e.g., Ref. [6]. Several models have been introduced afterward (see, e.g., the discussions in Refs. [8–10]). Sokolov *et al.* [11–13] expressed the radiation backreaction of the electron motion in terms of the emission probability. They succeeded in deriving the RR force experienced by an electron from QED principles, extending the applicability to QED strong fields. For relatively weak fields, the Landau-Lifshitz (LL) equation is widely accepted. It does not produce inconsistencies (such as exponentially diverging acceleration even without an external field) and it is equivalent to the Lorentz-Abraham-Dirac equation up to first order in the coupling parameter [10]. By perturbation theory, the LL form was derived from basic considerations of energy and momentum conservation [10,14]. The limits of the LL radiation force applicability in terms of wave amplitude and frequency was discussed by Bulanov *et al.* [15]. Most recently, Seto *et al.* [16] introduced a nonaveraged radiation balance rule to derive the proper nonrelativistic limit. In the ultrarelativistic limit the LL form was confirmed.

Relativistic motion and radiation of an electron in the field of a (single) intense laser pulse was considered by several authors [10,17–19]. Strong effects of radiation damping were found via relativistic particle-in-cell simulations [17]. It was detected that because of radiation damping, an initially immobile electron does not move along the so-called figure-8 trajectories in the field of a linearly polarized laser pulse [18]. Based on analytical solutions, the characteristic radiation patterns were obtained [10]. In contrast to the dissipationless situation, with a radiation reaction an electron can gain energy when it is overrun by a plane laser pulse [19].

A different situation appears for the coherent motion of electrons in two colliding laser pulses that are slightly detuned in frequency. Stochastic motion of electrons occurs when the amplitudes of the two laser pulses exceed some threshold [20]. One of the early papers on this subject by Mendonça and Doveil [21] demonstrated the stochastic motion of electrons in two electromagnetic waves and explained the phenomenon by the stochastic instability near the separatrices. Particle-in-cell simulations by Sentoku *et al.* [22] were used to identify the key process leading to stochastic heating. It was argued [23] that dynamical chaos can be made responsible for classical diffusion acceleration.

The dissipative nature of radiation reaction will definitely change the so far Hamiltonian nature of chaos at laser intensities exceeding 10^{23} W cm⁻². Indeed, it has been found that the radiation reaction force leads to a significant electron cooling [24]. This effect can be associated with the contraction of the available phase-space volume [25,26]. We enter the area of dissipative chaos with its many new phenomena such as (strange) attractors. They will become important for accelerated electrons [2,3] as well as for many other applications [9,27,28].

In the present paper we investigate phase-space contraction for the motion of electrons in high-intensity laser fields. First, to better understand the more complex two-pulse results, we start with the motion of an electron in a single electromagnetic wave. Changes of the phase space due to a radiation reaction are demonstrated; the effect of the radiation reaction is smaller

for propagation in the same direction as the plane wave compared to counterpropagation [24]. Second, in the case of two colliding laser beams, when stochastic heating can occur, it is shown that the radiation reaction causes attractors. In certain parameter regimes, electron motion converges to regular attractors. Stochastic heating is significantly reduced. The forms of the attractors are presented and the regions of appearance of regular attractors are discussed. The results confirm the general prediction of phase-space contraction and provide quantitative results for the speed of contraction as well as the type of attracted motion.

II. RADIATION FORCE AND PHASE-SPACE CONTRACTION

Based on the current understanding of the radiation reaction, in the present paper we use the LL form for the radiation reaction to find the motion of electrons in ultrarelativistic laser fields. When one normalizes time with the inverse wave frequency ω^{-1} , i.e., $t\omega \rightarrow t$, velocity components of $\mathbf{v}$ by c , components of momentum $\mathbf{p}$ by mc , and uses the electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$ in dimensionless forms $\frac{e\mathbf{E}}{mc\omega} \rightarrow \mathbf{E}$ and $\frac{e\mathbf{B}}{mc\omega} \rightarrow \mathbf{B}$ corresponding to a dimensionless vector potential $\frac{e\mathbf{A}}{mc} \rightarrow \mathbf{a}$, the spatial components of the radiation force are [7,24,26]

$$\mathbf{F}_R = -\mathcal{F}\gamma \left[\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \right) \mathbf{E} + \mathbf{v} \times \left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \right) \mathbf{B} \right] + \mathbf{F}_{RR}, \quad (1)$$

where

$$\mathbf{F}_{RR} = \mathcal{F} \{ \mathbf{E}(\mathbf{v} \cdot \mathbf{E}) + \mathbf{E} \times \mathbf{B} + \mathbf{B} \times (\mathbf{B} \times \mathbf{v}) - \gamma^2 \mathbf{v} [(\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{E} \cdot \mathbf{v})^2] \}. \quad (2)$$

The (dimensionless) prefactor $\mathcal{F}$ is

$$\mathcal{F} = \frac{2}{3} \frac{e^2 \omega}{mc^3} \approx 1.474 \times 10^{-8} \quad (3)$$

for a $\lambda = 800$ nm wavelength ($\omega = 2\pi c/\lambda$).

Considering the various terms of Eq. (1), it is physically consistent to neglect the first term. The argument is that a particle with a spin degree of freedom is also subject to a force in an external field and that this force (the Frenkel force), in the case of a plane wave, is much larger than the first term in Eq. (1). Since we shall work with a classical equation of motion neglecting the spin, it is consistent to drop the first term and keep only $\mathbf{F}_{RR}$.

The vector-analytical relations $\mathbf{B} \times (\mathbf{B} \times \mathbf{v}) = (\mathbf{B} \cdot \mathbf{v})\mathbf{B} - \mathbf{B}^2\mathbf{v}$, $\mathbf{E} \cdot (\mathbf{v} \times \mathbf{B}) = -\mathbf{v} \cdot (\mathbf{E} \times \mathbf{B})$, and $(\mathbf{v} \times \mathbf{B})^2 = \mathbf{v}^2\mathbf{B}^2 - (\mathbf{v} \cdot \mathbf{B})^2$ are helpful in determining the strength and direction of the radiation force. For example, when starting with particles counterpropagating to the plane wave (the unit wave number is $\hat{k}$), we get the *initial* force

$$\mathbf{F}_{RR}^0 = \mathcal{F} \frac{1+v_0}{1-v_0} |\mathbf{E}|^2 \hat{k} \quad \text{for } \mathbf{v} = -v_0 \hat{k}. \quad (4)$$

In contrast, when the particles initially propagate in the same direction as the plane wave, one obtains

$$\mathbf{F}_{RR}^0 = \mathcal{F} \frac{1-v_0}{1+v_0} |\mathbf{E}|^2 \hat{k} \quad \text{for } \mathbf{v} = v_0 \hat{k}. \quad (5)$$

Clearly, the value for counterpropagation is larger than for copropagation (when the same magnitude v_0 is assumed for both cases).

With respect to phase-space contraction, Tamburini *et al.* [24] reported the important result

$$\nabla_{\mathbf{p}} \cdot \mathbf{F}_{RR} = -\mathcal{F} \left\{ 2 \frac{\mathbf{E}^2 + \mathbf{B}^2}{\gamma} + 4\gamma [(\mathbf{v} \times \mathbf{E})^2 + (\mathbf{v} \times \mathbf{B})^2 - 2\mathbf{v} \cdot (\mathbf{E} \times \mathbf{B})] \right\} \leq 0, \quad (6)$$

which expresses phase-space contraction. The first part of Eq. (6), i.e., the expression for the divergence, immediately follows from a direct calculation of $\nabla_{\mathbf{p}} \cdot \mathbf{F}_{RR}$ making use of $\partial\gamma/\partial p_i = \gamma^{-1} p_i$ and $\partial\gamma^{-1}/\partial p_i = -\gamma^{-3} p_i$.

When discussing the sign [24] of the term

$$A \equiv \frac{\mathbf{E}^2 + \mathbf{B}^2}{2\gamma^2} + (\mathbf{v} \times \mathbf{E})^2 + (\mathbf{v} \times \mathbf{B})^2 - 2\mathbf{v} \cdot (\mathbf{E} \times \mathbf{B}), \quad (7)$$

we may use (in dimensionless variables) for the electromagnetic wave

$$\mathbf{E} \perp \mathbf{B}, \quad |\mathbf{E}| = |\mathbf{B}|, \quad \gamma^2 = \frac{1}{1 - \mathbf{v}^2}. \quad (8)$$

The vectors $\mathbf{E}$, $\mathbf{B}$, and $\mathbf{E} \times \mathbf{B}$ define three orthogonal directions. Next, we decompose the velocity accordingly into orthogonal components

$$\mathbf{v} = \mathbf{v}_E + \mathbf{v}_B + \mathbf{v}_{E \times B}, \quad \mathbf{v}^2 = \mathbf{v}_E^2 + \mathbf{v}_B^2 + \mathbf{v}_{E \times B}^2. \quad (9)$$

Introducing that into

$$A = \frac{\mathbf{E}^2 + \mathbf{B}^2}{2} (1 - \mathbf{v}^2) + \mathbf{v}^2 \mathbf{E}^2 - (\mathbf{v} \cdot \mathbf{E})^2 + \mathbf{v}^2 \mathbf{B}^2 - (\mathbf{v} \cdot \mathbf{B})^2 - 2\mathbf{v} \cdot (\mathbf{E} \times \mathbf{B}) \quad (10)$$

leads to

$$A = [\mathbf{v}^2 + 1 - \mathbf{v}_E^2 - \mathbf{v}_B^2 - 2v_{E \times B}^2] |\mathbf{E}|^2 \quad (11)$$

or

$$A = [v_{E \times B}^2 + 1 - 2v_{E \times B}] |\mathbf{E}|^2 = (v_{E \times B} - 1)^2 |\mathbf{E}|^2 \geq 0. \quad (12)$$

This shows the right-hand-side of Eq. (6).

Following Tamburini *et al.* [24], the available phase-space volume contracts according to

$$\frac{dS}{dt} = \int d^3q d^3p f \nabla_{\mathbf{p}} \cdot \mathbf{F}_{RR} \leq 0, \quad (13)$$

where $S = -\int d^3q d^3p f \ln f$ is the entropy associated with a distribution of particles with the distribution function $f = f(\mathbf{q}, \mathbf{p}, t)$ in phase space. Note that this result was obtained for a bunch of electrons interacting with a plane wave where collective fields are assumed to be negligible compared with the plane-wave fields [24]. This effectively means that the contraction of the phase space was proven for systems of noninteracting electrons. In the latter case we can use the results for the ensemble dynamics when we consider the behavior of a single system (which is characterized by the following by Poincaré plots).

Again, counterpropagation can be compared with copropagation. We have

$$\nabla_{\mathbf{p}} \cdot \mathbf{F}_{\text{RR}} = -4\mathcal{F}|\mathbf{E}|^2 \left[2\gamma_0 v_0(1 + v_0) + \frac{1}{\gamma_0} \right] \text{ for } \mathbf{v} = -v_0 \hat{\mathbf{k}}, \quad (14)$$

i.e., counterpropagation. In contrast, for copropagation

$$\nabla_{\mathbf{p}} \cdot \mathbf{F}_{\text{RR}} = -4\mathcal{F}|\mathbf{E}|^2 \frac{1}{(1 + v_0)^2 \gamma_0^3} \text{ for } \mathbf{v} = v_0 \hat{\mathbf{k}}. \quad (15)$$

Clearly, for $\gamma_0 \gg 1$ the phase-space contraction should be much faster for counterpropagation than for copropagation. Equations (14) and (15) show that in any case the characteristic time for phase-space contraction scales with $\mathcal{F}|\mathbf{E}|^2$.

III. MOTION IN A SINGLE WAVE

We start with the most simple situation of electrons in a single, plane electromagnetic wave. We integrate the equation of motion using a standard Runge-Kutta method. Without a radiation reaction, we only take into account the electric and Lorentz force,

$$\frac{d\mathbf{p}}{dt} = \mathbf{F}_L = -(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (16)$$

In the other case, with radiation damping, the equation of motion is

$$\frac{d\mathbf{p}}{dt} = \mathbf{F}_L + \mathbf{F}_{\text{RR}}. \quad (17)$$

The laser fields are assumed to be linearly polarized in the y direction. The associated dimensionless vector potential is

$$\mathbf{a} = a_0 \cos(x - t) \hat{\mathbf{e}}_y. \quad (18)$$

The velocities of the electrons are characterized as comoving when being in the wave propagation direction and counter-moving in the opposite case. We compare the two cases, with and without a radiation reaction, for the situation when

all initial velocities are countermoving. That initial situation is purposely chosen to demonstrate the fast phase-space contraction for counterpropagation. (The corresponding case with positive p_{x0} was also calculated; we do not present the results here since they show the expected slower phase-space contraction.)

Figure 1(a) shows results from integration of Eq. (16) for electrons with different initial positions and momenta being (initially) evenly distributed within one period of the wave. The wave amplitude is $a_0 = 30$. All initial momenta p_{0x} are negative and in the range between $p_{0x,\text{min}} = -1$ and $p_{0x,\text{max}} = -100$. The trajectories in the lower part of Fig. 1(a) show that most particles keep their counterdirection, i.e., they propagate with velocities v_x against the propagation direction of the wave. Only a very few electrons are pushed by the wave into the codirection.

In order to study the phase space of the motion we render Poincaré section plots. Defining $\eta = x \bmod 2\pi$, we plot the longitudinal momentum p_x versus η at times $t = 2\pi n$, $n = 0, 1, \dots$. The Poincaré plot in Fig. 1(a) shows that a broad range of negative longitudinal momenta p_x is present in the system. The motion is regular and represents the relativistic behavior that has been reported in many papers [29].

The Poincaré section plot in Fig. 1(a) is easy to understand. The plane electromagnetic wave depends on only the combination $x - t$ and shows no dependence on the transverse coordinates. From these symmetries the conservation laws $\mathbf{p}_{\perp} = \mathbf{a}$ and $E - p_x = \gamma_0 - p_{x0}$ follow [19]. The latter relation can be written as

$$\gamma - \gamma_0 = p_x - p_{x0}, \quad (19)$$

where

$$\gamma_0 = \sqrt{1 + p_{x0}^2}, \quad \gamma = \sqrt{1 + p_x^2 + \mathbf{p}_{\perp}^2} = \sqrt{1 + p_x^2 + \mathbf{a}^2}. \quad (20)$$

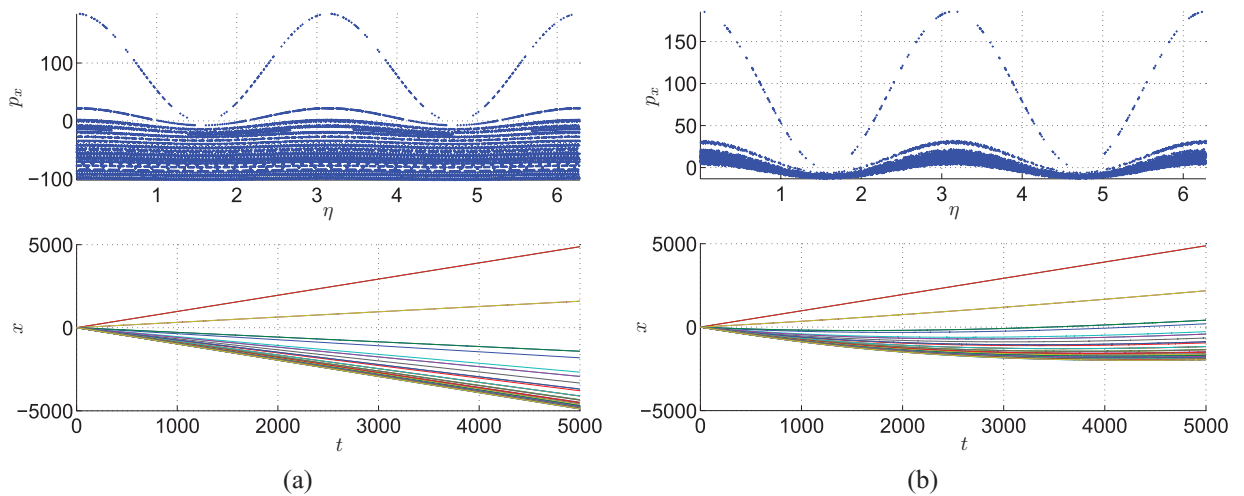


FIG. 1. (Color online) Poincaré section plot p_x vs η (top) and longitudinal position x vs t (bottom) for (a) electrons moving under the action of $\mathbf{F} = \mathbf{F}_L$, i.e., without a radiation reaction, in a wave with $a_0 = 30$. Colors in the bottom graph represent different initial positions and momenta. (b) Same as (a) but including the radiation force, i.e., $\mathbf{F} = \mathbf{F}_L + \mathbf{F}_{\text{RR}}$. The influence of the radiation force on counterpropagating particles is clearly visible. The Poincaré plot is generated from the trajectories in the bottom graph, but considers only motion for $t > 5000$.

From here

$$\gamma\gamma_0 = 1 + \frac{a^2}{2} + p_x p_{x0} \quad (21)$$

follows. Eliminating γ from Eqs. (19) and (21), we obtain

$$p_x = \frac{1 + \frac{a^2}{2}}{\gamma_0 - p_{x0}} - \gamma_0. \quad (22)$$

When p_{x0} is varied in the range $-100 \leq p_{x0} \leq -1$, only the small $|p_{x0}|$ values (corresponding to small γ_0 values) can lead to $p_x > 0$. For the majority of (initial) p_{x0} values p_x remains negative (counterpropagating).

When we take into account the radiation reaction force, i.e., integrating Eq. (17), we turn from a Hamiltonian system to a dissipative system. Integrating Eq. (17) for an ensemble of noninteracting electrons with the same initial conditions as in the previous example, we obtain the results shown in Fig. 1(b). From the bottom graph in Fig. 1(b) it can be seen that only initially the longitudinal motion of the particles is similar to the motion in the nondissipative system. After a certain time the particles start slowing down their motion against the propagation direction of the wave and eventually even obtain a positive momentum p_x . This behavior is also reflected in the Poincaré section of the phase space [top graph of Fig. 1(b)]. For the phase-space plot, we only present data after a certain time t to show the contraction in phase space. Dissipative effects need a certain time to enforce clear changes in the phase-space structure. In the case shown we started plotting the Poincaré section at $t = 5000$. The resulting plot is clearly different from the corresponding plot in Fig. 1(a). It demonstrates the contraction of the phase space due to the radiation force. We notice the absence of a broad range of negative momenta. This demonstrates that the particles initially moving against the wave will be influenced strongly by the radiation reaction force. The resulting phase-space structure is narrower than in the case of undamped motion.

IV. MOTION IN TWO COLLIDING WAVES

The problem of electron motion in two colliding waves is of utmost interest for stochastic heating [20]. While motion in a single wave is regular, under the action of two waves the dynamics may become chaotic. The increase in the random part of the velocity can be interpreted as heating. Here we examine the simplest case for stochastic laser heating [20], i.e., the motion of an electron in two colliding, linearly polarized waves. The total vector potential is assumed as

$$\mathbf{a}(x, t) = a_1 \cos(x - t) \hat{e}_y + a_2 \cos(x + t) \hat{e}_y. \quad (23)$$

For simplicity both waves have the same frequency. We have checked that a difference in frequencies of the two waves does not affect the main conclusions being presented next. In the context of stochastic heating, the right-traveling wave could play the role of the incident wave, while the counterpropagating wave is a reflected wave.

As a reference case, we first consider the motion of electrons in two colliding waves without taking into account radiation damping. Then the motion is of Hamiltonian character, just as for an electron moving in a single wave. The fundamental

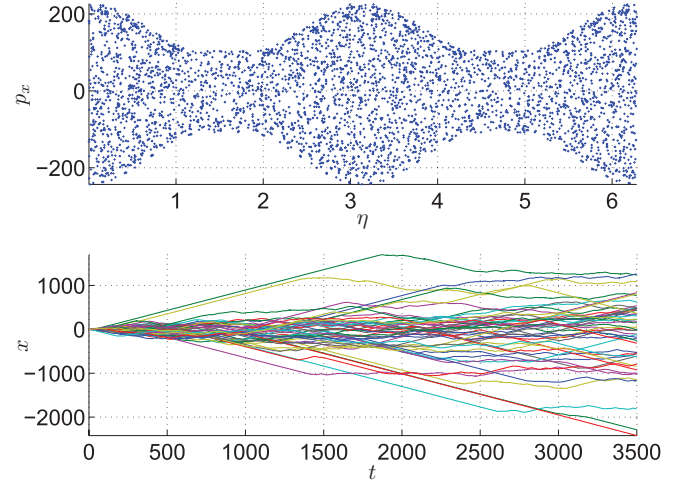


FIG. 2. (Color online) Poincaré section plot p_x vs η (top) and longitudinal position x vs t (bottom) for electrons moving due to $\mathbf{F} = \mathbf{F}_L$ in two waves with $a_1 = 60$ and $a_2 = 60$. Colors in the bottom graph represent different initial positions and momenta.

difference, however, is that now the system can exhibit chaotic behavior [20,21].

Figure 2 demonstrates stochasticity (Hamiltonian chaos) for the motion of electrons in two colliding waves with $a_1 = a_2 = 60$. In the bottom graph we plot the longitudinal space coordinate versus time for different initial conditions. The initial conditions are made up from different initial positions x_0 in the interval $[0, 2\pi]$ and several initial momenta p_{x0} between $p_{x0, \min} = -1$ and $p_{x0, \max} = 1$. The motion along the x axis becomes chaotic for all initial conditions considered. This is reflected in the Poincaré plot (top of Fig. 2), where we observe a global stochastic behavior. Electrons that initially have a rather small momentum may become strongly relativistic, which is in agreement with earlier predictions [23].

It should be mentioned that not all initial momenta lead to stochastic motion. In the top of Fig. 2 only the area of stochastic motion is shown. The stochastic area is bounded by (averaged) momenta values $\langle p_{x, \text{limit}}^{\text{Hamilton}} \rangle$ whose absolute values are less than or equal to 200. The magnitude of $\langle p_{x, \text{limit}}^{\text{Hamilton}} \rangle$ increases with the wave amplitude $a_1 (=a_2)$. A fit of numerical data leads to $\langle p_{x, \text{limit}}^{\text{Hamilton}} \rangle \sim \pm 3a_1$ for $a_1 \approx a_2$. We shall return to this point later.

Next we consider the same situation, but now we include radiation damping in the equation of motion. The system becomes dissipative and we switch from Hamiltonian chaos to dissipative chaos. Completely new dynamical phenomena, such as attractors, may appear. In the literature [24,25], contraction of the phase space has been predicted. The open question is the topological structure of the contracted phase space, i.e., whether attractors appear and if so, what the physical consequences of the attractors are. We exemplify answers to these questions in Fig. 3. When considering the same initial conditions for the electrons as in the calculations for Fig. 2, the motion of the electrons along the x axis now shows a very different character. In the beginning the motion still shows chaotic behavior. After some time, each electron ends up propagating with constant velocity in either the positive or the negative x direction. Irrespective of direction,

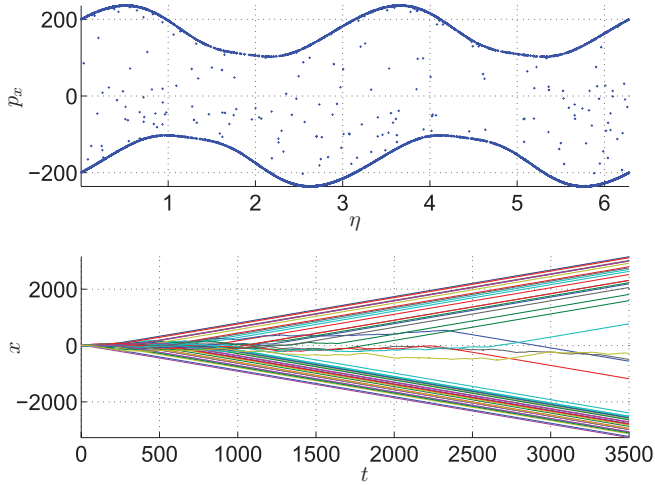


FIG. 3. (Color online) Same as Fig. 2, but for motion including the radiation force. The dissipation of energy by radiation leads to a phase-space contraction, resulting in the generation of two attractors (lines ending at $p_x = \pm 200$ in the Poincaré section). In the top graph the Poincaré plot considers values of the trajectories for only $t > 3500$.

their terminal speed approaches 1. The corresponding Poincaré section of phase space is shown in the top of Fig. 3. The points are generated by considering data of the trajectories after an initial phase; in this case we consider data for only $t > 3500$. The resulting Poincaré plot clearly shows the existence of two regular attractors $\langle p_{x \text{ limit}} \rangle \approx \pm 200$. The few remaining data points in the phase-space region between the two attractors originate from trajectories that have not yet completely converged onto one of the attractors. For later times, these points will disappear. To estimate $\langle p_{x \text{ limit}} \rangle$ we compare the parallel x component of the RR force

$$\mathbf{F}_{\text{RR}|x} = \mathcal{F}\{1 - v_x - \gamma^2 v_x [1 - v_x]^2\} |\mathbf{E}|^2 \quad (24)$$

with the parallel component of the Lorentz force $\mathbf{F}_L|_x \approx \gamma^{-1} |\mathbf{E}|^2$. Balancing occurs at the order

$$\langle p_{x \text{ limit}} \rangle^3 \sim \frac{1}{\mathcal{F}}. \quad (25)$$

We have numerically confirmed this rough estimate by (artificially) varying $\mathcal{F}$ and found the approximate proportionality factor $\frac{1}{4}$.

When comparing p_x with $\dot{x} = v_x$, we should have in mind that the γ factor in $p_x = \gamma v_x$ is determined by the total velocity $\mathbf{v}$, which also contains the quiver velocity in the y and z directions. Clearly, the motion of the electrons on the attractor with $|v_x| < 1$ and $|p_x| \sim O(200)$ is ultrarelativistic and the γ factor has grown significantly over its initial value. Of course, we do not have trapping in a single wave though since the phase velocity of the wave is $\omega/k = 1$.

The attractors presented here occur in a broad region of values a_1 and a_2 with $a_1 \approx a_2$. However, they do not persist with their regular structures for all $a_1 \approx a_2$. When $a_1 \approx a_2 \gtrsim 70$ the attractors turn into strange attractors. When varying the amplitude $a_1 \approx a_2$ we found the criterion

$$\langle p_{x \text{ limit}}^{\text{Hamilton}} \rangle \geq \langle p_{x \text{ limit}} \rangle \sim \frac{1}{\mathcal{F}^{1/3}} \quad (26)$$

for the appearance of strange attractors. The validity of the criterion (26) is demonstrated in Fig. 4. Figure 4(a) is for small $a_1 = a_2$ fulfilling $\langle p_{x \text{ limit}}^{\text{Hamilton}} \rangle < \langle p_{x \text{ limit}} \rangle$, while Fig. 4(b) shows the strange attractor for $\langle p_{x \text{ limit}}^{\text{Hamilton}} \rangle > \langle p_{x \text{ limit}} \rangle$. Note also that the boundaries of the stochastic regions in the top plots of Figs. 4(a) and 4(b) agree with the scaling $\langle p_{x \text{ limit}}^{\text{Hamilton}} \rangle \sim \pm 3a_1$ for $a_1 \approx a_2$.

So far we have considered two colliding waves with equal amplitudes. When the second wave is interpreted as a reflected part of the first wave, generally the reflected amplitude a_2 should be smaller than the amplitude a_1 . We found that as long as $a_1 \approx a_2$ the phase space will be very similar to the one shown in Figs. 3 and 4. However, when a_1 is much larger than a_2 , we observe only one attractor (with positive p_x) in phase

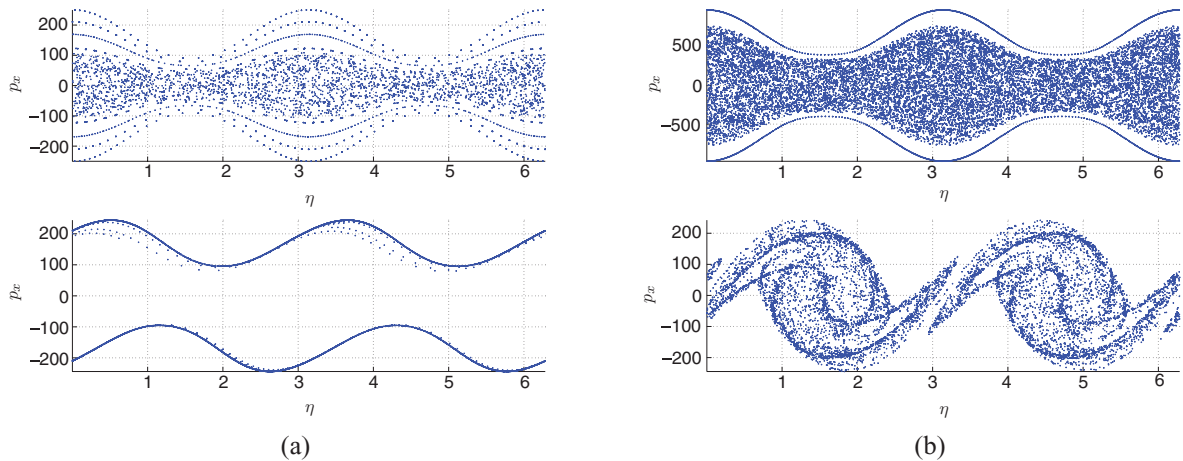


FIG. 4. (Color online) Poincaré plots for (a) $a_1 = a_2 = 30$ and (b) $a_1 = a_2 = 200$, respectively, compared to $a_1 = a_2 = 60$ in Figs. 2 and 3. The top plots are for the Hamiltonian system without a radiation reaction while the lower plots are with a radiation reaction. Plot (a) shows 100 trajectories initiated in the region $-120 \leq p_x \leq 120$ for 500 time steps starting at $t = 18000$. Plot (b) shows 100 trajectories initiated in the region $-400 \leq p_x \leq 400$ for 1000 time steps beginning at $t = 2000$.

space. The precise ratio a_1/a_2 at which the lower attractor will vanish depends on the absolute value of a_1 .

Another remark concerns the characteristic attraction time T . The numerical results are consistent with the estimate $T \sim [\mathcal{F}|\mathbf{E}|^2\gamma]^{-1}$, which follows from Eq. (14). The electrons have to stay long enough in the waves in order to approach the attractors. Thus the regular motion due to phase-space contraction will be observable only for colliding laser pulses with reasonable widths.

V. CONCLUSION

In this paper we considered electron motion in high-intensity laser fields. In this case the electron motion becomes ultrarelativistic and the electron dynamics will be influenced by a radiation reaction. It was previously [24] predicted that the radiation force should cause phase-space contraction. When considering the motion of an electron in a *single* electromagnetic wave, the effect of the radiation reaction is smaller for

propagation in the same direction as the plane wave compared to counterpropagation. In this work we have demonstrated numerically that the phase-space contraction occurs and how it affects stochastic heating. We mainly considered the case of *two colliding laser beams*, when stochastic heating may occur. It was shown that a radiation reaction causes attractors. In the parameter regimes considered, electron motion converges to regular attractors. Stochastic heating is significantly reduced. The forms of the attractors were depicted and rough criteria for the existence of different forms of attractors were presented. The results confirm the general prediction of phase-space contraction and provide quantitative results for the speed of contraction as well as the type of attracted motion.

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- [1] G. A. Mourou, T. Tajima, and S. V. Bulanov, *Rev. Mod. Phys.* **78**, 309 (2006).
 - [2] G. Malka and J. L. Miquel, *Phys. Rev. Lett.* **77**, 75 (1996).
 - [3] G. V. Stupakov and M. S. Zolotarev, *Phys. Rev. Lett.* **86**, 5274 (2001).
 - [4] A. Di Piazza, K. Z. Hatsagortsyan, and C. H. Keitel, *Phys. Rev. Lett.* **102**, 254802 (2009).
 - [5] C. Harvey and M. Marklund, *Phys. Rev. A* **85**, 013412 (2012).
 - [6] P. A. M. Dirac, *Proc. R. Soc. London Ser. A* **167**, 148 (1938).
 - [7] L. D. Landau and E. M. Lifshitz, *The Classical Theory of Fields* (Pergamon, New York, 1994).
 - [8] F. Rohrlich, *Phys. Lett. A* **283**, 276 (2001).
 - [9] V. I. Berezhiani, R. D. Hazeltine, and S. M. Mahajan, *Phys. Rev. E* **69**, 056406 (2004).
 - [10] Y. Hadad, L. Labun, J. Rafelski, N. Elkina, C. Klier, and H. Ruhl, *Phys. Rev. D* **82**, 096012 (2010).
 - [11] I. V. Sokolov, *JETP* **109**, 207 (2009).
 - [12] I. V. Sokolov, N. M. Naumova, J. A. Nees, G. A. Mourou, and V. P. Yanovsky, *Phys. Plasmas* **16**, 093115 (2009).
 - [13] I. V. Sokolov, J. A. Nees, V. P. Yanovsky, N. M. Naumova, and G. A. Mourou, *Phys. Rev. E* **81**, 036412 (2010).
 - [14] S. E. Gralla, A. I. Harte, and R. M. Wald, *Phys. Rev. D* **80**, 024031 (2009).
 - [15] S. V. Bulanov, T. Z. Esirkepov, M. Kando, J. K. Koga, and S. S. Bulanov, *Phys. Rev. E* **84**, 056605 (2011).
 - [16] K. Seto, H. Nagatomo, J. Koga, and K. Mima, *Phys. Plasmas* **18**, 123101 (2011).
 - [17] A. Zhidkov, J. Koga, A. Sasaki, and M. Uesaka, *Phys. Rev. Lett.* **88**, 185002 (2002).
 - [18] A. L. Galkin, V. V. Korobkin, M. Y. Romanovsky, and O. B. Shiryayev, *Quantum Electron.* **37**, 903 (2007).
 - [19] G. Lehmann and K. H. Spatschek, *Phys. Rev. E* **84**, 046409 (2011).
 - [20] Z. M. Sheng, K. Mima, Y. Sentoku, M. S. Jovanovic, T. Taguchi, J. Zhang, and J. Meyer-ter-Vehn, *Phys. Rev. Lett.* **88**, 055004 (2002).
 - [21] J. Mendonça and F. Doveil, *J. Plasma Phys.* **28**, 485 (1982).
 - [22] Y. Sentoku, V. Y. Bychenkov, K. Flippo, A. Maksimchuk, K. Mima, G. Mourou, Z. M. Sheng, and D. Umstadter, *Appl. Phys. B* **74**, 207 (2002).
 - [23] V. S. Rastunkov and V. P. Krainov, *Laser Phys.* **15**, 262 (2005).
 - [24] M. Tamburini *et al.*, *Nucl. Instrum. Methods Phys. Res. Sect. A* **653**, 181 (2011).
 - [25] R. D. Hazeltine and S. M. Mahajan, *Phys. Rev. E* **70**, 046407 (2004).
 - [26] M. Tamburini, F. Pegoraro, A. DePiazza, C. H. Keitel, and A. Macchi, *New J. Phys.* **12**, 123005 (2010).
 - [27] F. Andersson, P. Helander, and L. G. Eriksson, *Phys. Plasmas* **8**, 5221 (2001).
 - [28] R. D. Hazeltine and S. M. Mahajan, *Phys. Rev. E* **70**, 036404 (2004).
 - [29] A. Pukhov, *Rep. Prog. Phys.* **66**, 47 (2003).